Ex 1:

a) $\lim_{n\to\infty}\underbrace{\left(\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+\ldots+\frac{1}{\sqrt{n^2+n}}\right)}_{S}$

$$\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+1}}+\ldots+\frac{1}{\sqrt{n^2+1}} \le S \le \frac{1}{\sqrt{n^2+n}}+\ldots+\frac{1}{\sqrt{n^2+n}}$$

$$\frac{n}{\sqrt{n^2+1}} \le S \le \frac{n}{\sqrt{n^2+n}} \Rightarrow \lim \frac{n}{n\sqrt{1+\frac{1}{n^2}}} \le \lim S \le \lim \frac{n}{n\sqrt{1+\frac{1}{n}}}$$

$\lim \frac{n}{n\sqrt{1+\frac{1}{n^2}}} = 1$, $\lim \frac{n}{n\sqrt{1+\frac{1}{n}}} = 1$

$\Rightarrow \lim S = 1$

b) $\lim\left(\frac{1}{n^2+1}+\frac{2}{n^2+2}+\ldots+\frac{n}{n^2+n}\right)$

$$\frac{1}{n^2+n} \le \frac{1}{n^2+1} \le \frac{1}{n^2+1}$$

$$\frac{2}{n^2+n} \le \frac{2}{n^2+1} \le \frac{2}{n^2+1}$$

$$\frac{n(n+1)}{2(n^2+n)} \le S \le \frac{n(n+1)}{2(n^2+1)}$$

$\lim \frac{n(n+1)}{2(n^2+n)} = \frac{1}{2}$, $\lim \frac{n(n+1)}{2(n^2+1)} = \frac{1}{2}$

c) $\lim \frac{[x]+[2x]+\ldots[nx]}{n^2}$:

d) $x_n = \sum_{k=2}^{n} k\frac{1}{\sqrt{n^k+n+1}}$

$$\frac{n}{\sqrt[n]{n^n+n+1}} \le S \le \frac{n}{n+n+1}$$

$$\downarrow \qquad \downarrow$$

$$1 \qquad 1$$

e) $$\sum_{k=1}^{n} \sqrt{1+\frac{k}{n^2}} - n = \sum_{k=1}^{n} \frac{\sqrt{n^2+k}-n^2}{n}$$

$$= \frac{\sqrt{n^2+k}}{n} - n = \frac{\sqrt{n^2+k-n^2}}{n}$$ ✓

$$\frac{\sqrt{n^2+k}-n}{n} = \frac{k}{n(\sqrt{n^2+k}+n)}$$

$$\frac{1}{\sqrt{n^2+1}+n} \le \frac{k}{n(\sqrt{n^2+k}+n)} \le \frac{1}{1+\frac{k}{n^2}-n} =$$

$$\frac{\sqrt{n^2+k}-n^2}{n} \le 2$$!

Ex 2) a) $$\lim \frac{1^p+2^p+\dots+n^p}{n^{p+1}} = \frac{(n+1)^p}{(n+1)^{p+1}-n^{p+1}}$$

$$= \frac{1}{(n+1)-\left(\frac{n}{(n+1)}\right)}$$?

b) $$x_n = \frac{1}{\ln n}\left(1+\frac{1}{2}+\dots+\frac{1}{n}\right)$$

$$\frac{\frac{1}{n+1}}{\ln\frac{n+1}{n}} = \frac{1}{(n+1)\ln\left(n+\frac{1}{n}\right)} = 1$$ ✓

c) $\dfrac{\sqrt[n+1]{a}-1}{\ln\left(\frac{n+1}{n}\right)} = \dfrac{\sqrt[n]{a}-1}{\ln\left(1+\frac{1}{n}\right)\cdot\frac{1}{n}} =$

$\dfrac{\sqrt[n+1]{a}-1}{\frac{1}{n}} = \dfrac{a^{\frac{1}{n+1}}-1}{\frac{1}{n}} = \ln a$

Ex 3) a) $\dfrac{(n+1)!\,(n+1)^{n+1}}{(2n+2)!\,8^{n+1}} \cdot \dfrac{(2n)!\cdot 8^n}{n!\,n^n}$

$= \dfrac{(n+1)\cdot(n+1)}{(2n+1)(2n+2)\cdot 8} = \dfrac{n^2}{32n^2} = \dfrac{1}{32}$

b) $\dfrac{n}{\sqrt[n]{n!}} = \dfrac{(n+1)^{n+1}}{(n+1)!}\cdot\dfrac{n!}{n^n}$

$= \left(\dfrac{n+1}{n}\right)^n = e$

c) $\sqrt[n^2]{(pn!)} = (pn!)^{\frac{1}{n^2}} = \dfrac{\ln((pn)!)}{n^2}$

$= \dfrac{\ln((pn+p)!) - \ln((pn)!)}{(n+1)^2 - n^2}$

$= \ln\left(\dfrac{(pn+1)\cdot(pn+2)\cdot\ldots\cdot(pn+p)}{n+1}\right)$

$=$

a_n cresc

Ex 4) $a_{n+1} = a_n + \frac{1}{a_n}$ $\quad a_{n+1} - a_n = \frac{1}{a_n}$

$\ell = \ell + \frac{1}{\ell} \Rightarrow \frac{1}{\ell} \neq 0$, $\ell \geq 0 \Rightarrow a_n \to \infty$

$a_{n+1}^2 = a_n^2 + 2 + \frac{1}{a_n^2}$

$a_{n+1}^2 - a_n^2$

$\lim \frac{a_n^2}{n} = \lim \frac{a_{n+1}^2 - a_n^2}{} = 2 + \frac{1}{\ell^2} = 2$

$\Rightarrow \ell = \sqrt{2}$

Ex 5) $a_{n+1} = \operatorname{sin} a_n$

$\ell \sqrt{n} \cdot a_n =$ $\quad a_{n+1} - a_n = \operatorname{sin} a_n - a_n$

$\frac{a_{n+1}}{a_n} = \frac{\operatorname{sin} a_n}{a_n}$ $\quad 0 \leq a_n \leq \frac{\pi}{2}$ sin

$0 \leq \operatorname{sin} a_n \leq 1$

a_n

$-\frac{\pi}{2} \leq -a_n \leq 0$

$-\frac{\pi}{2} \operatorname{sin} - a_n \leq 1$

Ex 8) a) $1 + \frac{1}{2} + \ldots + \frac{1}{n} - \ln n$

a) ?

Ex a) 4.

Ex 10) a) $\lim \frac{1}{n+1} + \ldots + \frac{1}{2n}$

$= \left(1 + \frac{1}{2} + \ldots + \frac{1}{n+1} + \ldots + \frac{1}{2n} - \ln 2n\right) - \left(1 + \frac{1}{2} + \ldots + \frac{1}{n} - \ln n\right) + \ln 2n - \ln n$

$= \gamma_{2n} - \gamma_n + \ln 2 = \ln 2$

5) $\lim_{n\to\infty} n\left(\left(\frac{1}{1} + \dots + \frac{1}{2n} - \ln 2n\right) - \left(1 + \dots + \frac{1}{n} - \ln n\right) + \ln 2n - \ln n - \ln 2\right) =$

$$\frac{\frac{1}{2n+2}}{\frac{1}{n+1} - \frac{1}{n}} = \frac{\frac{1}{2n+2}}{\frac{n-n-1}{n^2+n}} = \frac{n^2+n}{-2}$$

$\gamma_{2n} - \gamma_n - \ln 2$?

✓: